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The British Tertiary igneous province: palaeomagnetism of the Arran dykes

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Summary. Five hundred and sixteen sites in Arran were sampled. After cleaning by heating and/or alternating fields 87 per cent yielded directions with $\alpha_{95} < 10^{\circ}$, comprising 435 dykes, 12 sills and one granite; polarities could be determined for some of the remainder. Directions are thought to be primary and on average are shallower than corresponds to the mean centred axial dipole, with many shallow reversely-magnetized dykes present.

Seventeen per cent of the dykes are normally magnetized, 5 per cent have intermediate directions. Polarity does not correlate with petrography, thickness or trend, but does depend upon location. In particular, on the NE coast up to 70 per cent are normal, and possibly belong to a different swarm.

Normal polarities were found for both sites in the Northern Granite, for Holy Island and for some of the sills. In conjunction with the known stratigraphy this shows that the rocks were formed during an R–N–R sequence of polarities, or more probably R–N–R–N. Using the radiometric age of 58 Ma for the Northern Granite and the best estimate of the polarity timescale the duration of activity was probably in the range 0.5 to 3.4 Ma if three polarity periods were involved, and 3.4 to 4.4 Ma if four were involved.

1 Introduction

This paper reports the results of a palaeomagnetic investigation of the dykes in Arran which is part of an intensive study of the whole British Tertiary igneous province (BTip) (Wilson 1970; Ade-Hall *et al.* 1972; Wilson, Dagley & Ade-Hall 1972a; Wilson *et al.* 1974; Mussett, Dagley & Eckford 1976 and Hall, Wilson & Dagley 1977). One of the most striking features of the igneous province is the predominant NW–SE trend of the dyke swarms and of these the Arran swarm is particularly outstanding for the density of intrusion and amount of dilation.

The dykes are readily seen along the shore and the collection (code numbers A03P001–A03P551) was made exclusively at coastal exposures along the south and east coasts from

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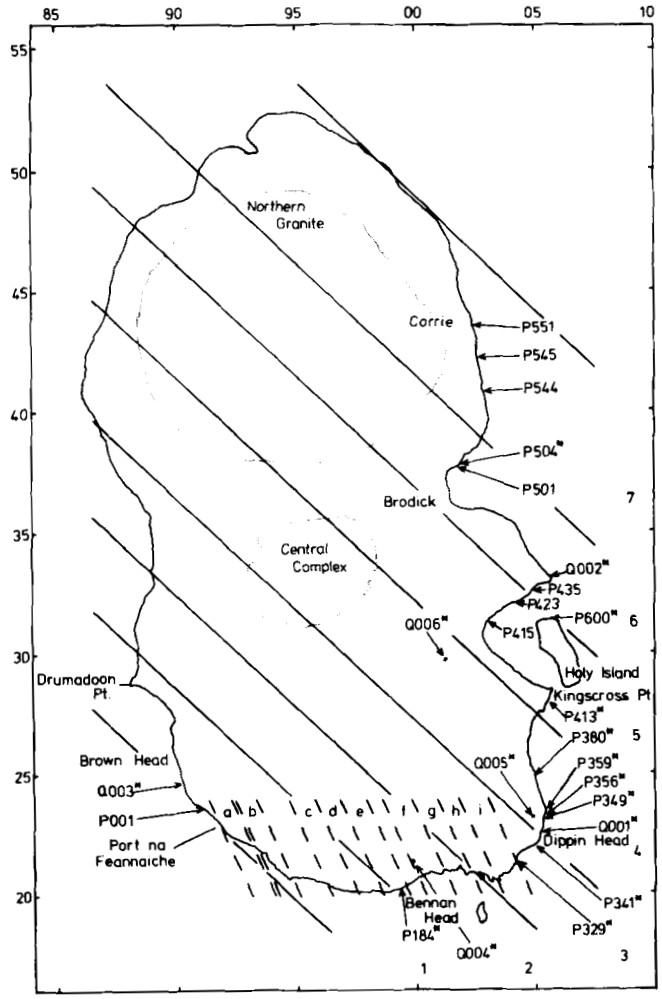


Figure 1. Outline map of Arran to show the sampling locations and the subdivisions used by Tyrrell and Knaap. Sheets and sills are indicated by an asterisk. The marginal numbering indicates location relative to the National Grid reference system.

just south of Brown Head (903248) to Corrie (025435) (OS 1 in. map 66) as indicated on the outline map (Fig. 1). In addition other bodies (certain sills and the Northern Granite – code numbers A03Q001–A03Q008) were sampled at various places inland, while offshore the Holy Island sill (A03P600) was sampled.

In all, samples were collected from 516 sites comprising 498 dykes, 16 identifiable sills and two granite bodies.

Two earlier studies have been made of the distribution of the dykes. Tyrrell (1928) made a statistical study in which he divided the island into seven adjoining NW–SE bands (1–7) each about 2 miles wide (Fig. 1), and among other things noted the trend and thickness of the dykes in each band. For comparison the dykes of the present study have been divided into the same groupings as closely as the boundaries can be identified (Table 1). Knaap (1973) in his study of the form and structure of the dykes divided the south coast into nine bands (a–i) parallel to 340° E, each 1 km wide (Fig. 1). Some short stretches of coast are not included in Knaap's bands but as these gaps are parts of coast with very little

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Table 1. Subdivision of dykes into Tyrrell and Knaap bands.

Tyrrell bands	Knaap bands	Our count/ Knaap's count	(% N)	(% R)	(% I)
(1) P001–P179	a P003–P025	24/26	9.5	90.5	0.0
Our count/Tyrrell count	b P028–P059	36/39	9.0	88.0	3.0
188/132	c P070–P100	33/41	25.0	72.0	3.0
16.6% N, 79.0% R; 4.3% I	d P101–P140	42/59	22.0	69.0	9.0
	e P141–P184	46/43	19.4	77.7	2.8
(2) P180–P288	f P186–P236	55/48	10.2	79.5	10.2
125/84	g P237–P269	34/52	8.8	88.0	2.9
8.4% N; 82.2% R; 9.3% I	h P270–P290	22/26	5.5	72.2	22.2
(3) P289–P348	i P291–P230	30*/56	0.0	96.3	3.5
0% N; 96.06% R; 4% I					
(4) P349–P410					
60/69					
7.7% N; 88.4% R; 3.9% I					
(5) P411–P435					
16*/49					
0% N; 90.0% R; 10% I					
(6) P502–P527					
27*/75					
72.0% N; 28.0% R; 0% I					
(7) P528–P550					
25*/51					
71.5% N; 23.8% R; 4.7% I					

* Incomplete.

exposure, correlation of our numbering with his bands is possible with fair agreement as to the number of dykes in each band (Table 1).

2 Collection and preparation

Four cores, 25-mm diameter, were taken from each unit using portable drilling equipment. As far as possible one sample was taken at the cooled edge, a second at ⅛ of the dyke width, a third at ¼ of the dyke width from the edge and the fourth on the mid line. The thickness of the dyke and the positions of each core were noted wherever possible but no systematic attempt was made to measure either the trend or tilt (hade) of the dyke for although in some cases the dykes formed prominent features with clear trends, the local trends and tilts of a dyke are very variable (see Knaap 1973): often there was insufficient exposure to measure one or the other. In most cases the tilt was considered to be original.

Each core was oriented by means of timed Sun-sights or geographic markers: no cores were oriented using a magnetic compass. The accuracy of the watches used for timing were checked each day with BBC time signals. One inch OS maps were used to give the geographically-determined bearings. The overall precision is estimated to be 3° or better.

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All the locations were marked on aerial photographs at the time of collection and later copied on to 6 in. maps.

In the laboratory each core was cut to provide at least one 25-mm length cylinder for magnetic measurements and a thin section for microscopic examination.

3 Measurement and analysis

The palaeomagnetic samples were measured using a parastatic magnetometer and all samples were cleaned using alternating field demagnetization. Demagnetizations were carried out in 5 mT steps until it was clear that either a stable direction had been reached or that the sample had become too weak or the successive directions too random to justify further treatment. Thermal demagnetizations were carried out on additional samples for 12 of the dykes; one each from Tyrrell bands 3 and 5, two each (one normal, one reversed) from the others.

Mean directions for each unit were found by combining the directions from the four magnetically cleaned cores in the manner described by Dagley & Ade-Hall (1970). The individual results and mean directions have been thoroughly inspected to eliminate any anomalies which may arise through misorientation or mismeasurement. Some individual core data were rejected on the grounds that the core gave a quite different direction to the rest of the unit.

Magnetic susceptibility was also measured for each core using a Bison mutual inductance bridge, and Curie points have been determined for 128 dykes.

A preliminary petrographic classification of the thin sections has been made in this department and a more detailed investigation is being carried out by Dr T. J. Halsall (University of Reading).

4 Results and discussion

A stereogram of all the unit mean directions is shown in Fig. 2. We will first consider their individual validity and the possible effect of secondary processes, remagnetization and regional tilting, then go on to discuss their significance.

4.1 PRECISION OF THE UNIT MEAN DIRECTIONS

Fig. 3 shows a normalized histogram of $\alpha_{95} < 30^\circ$ for 500 units forming the A03P set; seven units having $\alpha_{95} > 30^\circ$ have been excluded. Only 10 units have α_{95} between 20 and 30° but 47 (i.e. ~ 10 per cent) have values between 10 and 20°; 60 per cent have $\alpha_{95} < 5^\circ$. This distribution of directions is improved slightly if units with intermediate directions are excluded (see Table 2) but is not greatly altered if units for which $\alpha_{95} > 20^\circ$ or even $\alpha_{95} > 10^\circ$ are excluded. Mean directions for the groups consisting of: (a) all of the dykes, (b) all excluding those with intermediate directions, (c) all the normal dykes and (d) all the reversed dykes were calculated for the conditions $\alpha_{95} < 20^\circ$ and $\alpha_{95} < 10^\circ$ and the results for each class were found to be indistinguishable between the two sets (Table 2). The distribution, then, is not the result of poorly-defined directions of individual dykes. However, in considering the significance of the data only those units for which $\alpha_{95} < 10^\circ$ have been used. A total of 448 units (87 per cent; 435 dykes, 12 sills, one granite) were acceptable at this level from the combined A03P and A03Q sections.

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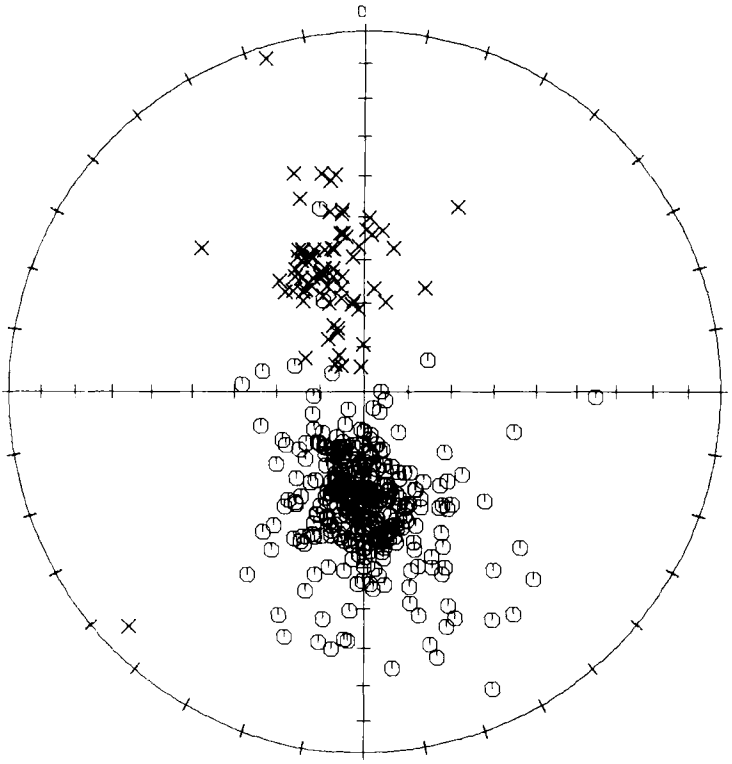


Figure 2. Equal-area projection of all unit mean directions. × – lower hemisphere, inclinations positive; ⏀ – upper hemisphere, inclinations negative.

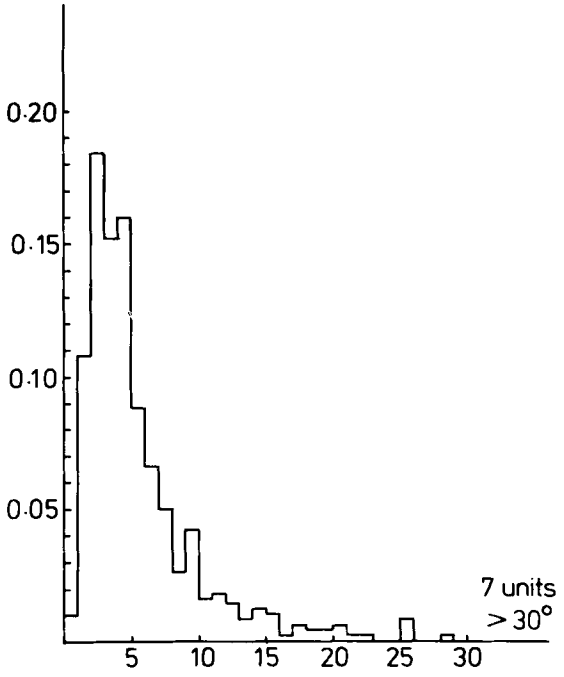


Figure 3. Normalized histogram of α_{95} for the dykes of P section.

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Table 2. Summary of mean directions.

	N	*R*	*D* (° ETN)	*I* (° down)	α (°)	δ (°)	Pole latitude	Pole longitude
Centred axial-dipole field	–	–	0	71°	–		90°	–
All P section	507	475.8	357.9	64.0	1.60	20.2	80.2	183.7
P section, $\alpha < 20°$	490	465.0	357.7	64.1	1.47	18.4	80.3	184.6
P section, $\alpha < 10°$	443	423.6	357.6	64.4	1.43	17.0	80.7	185.3
All P, N + R	454	441.1	359.0	64.6	1.13	13.7	80.9	179.4
P, N + R, $\alpha < 20°$	449	436.3	359.0	64.6	1.13	13.7	81.0	179.3
P, N + R, $\alpha < 10°$	417	405.5	359.1	65.0	1.16	13.5	81.6	179.3
All P, N	85	82.9	347.3	61.9	2.47	12.8	75.2	213.6
P, N, $\alpha < 20°$	83	81.0	347.3	61.9	2.46	12.6	75.2	213.6
P, N, $\alpha < 10°$	74	72.2	346.4	62.4	2.60	12.6	75.3	217.0
All P, R	369	358.8	182.0	−65.1	1.23	13.5	81.4	165.9
P, R, $\alpha < 20°$	366	355.9	182.0	−65.1	1.24	13.5	81.6	165.7
P, R, $\alpha < 10°$	343	333.7	182.1	−65.4	1.27	13.3	82.1	164.7
Dykes $\alpha < 10°$								
all dykes	435	417.4	357.6	64.5	1.4	16.8	80.7	185.2
N + R dykes	413	401.8	359.0	65.2	1.2	13.4	81.7	179.8
N dykes	73	71.2	346.7	62.4	2.6	12.6	75.5	216.5
R dykes	339	331.0	181.9	−65.6	1.3	13.2	82.2	165.5
Other intrusives:								
P184 Olivine dolerite, Bennan Head			184.1	−33.5	1.7			
P329 Craignurite, Porta Leacach			reversed					
P341 Quartz–dolerite, Dippin Head			reversed					
P349 South of Largybeg point			163.4	−27.3	5.2			
P356 Felsite, Largybeg point			196.7	−42.4	2.2			
P359 Quartz–dolerite, north of Largybeg point			207.0	−54.7	5.1			
P380 Quartz–dolerite, Creag Dhubh			188.1	−56.2	4.4			
P413 South of Kingscross point			reversed					
P504			288.8	−79.1	5.4			
P600 Holy Island			331.7	57.1	4.1			
Q001 Crinanite, north of Dippin point			155.0	−49.0	4.5			
Q002			204.7	−50.8	3.1			
Q003 Quartz–porphyry, Brown Head			350.3	72.6	4.1			
Q004 Quartz–porphyry, near Bennan Head			3.0	72.6	1.4			
Q005 An Garradh, west of Port na Gaitten			intermediate (shallow reverse)					
Q006			195.2	−58.4	3.3			
Q007 Coarse, older granite			340.6	86.3	3.0			
Q008 Fine, younger granite			normal					

N = number of units.
R = length of the resultant of the N vectors.
D = mean declination measured east of true north.
I = mean inclination (positive down).
α = α_{95}, semiangle of cone of 95% confidence for mean direction.
δ = angular stand deviation, $\cos^{-1}/(R/N)$.

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4.2 ORIGINALITY OF DIRECTIONS OF MAGNETIZATION

4.2.1 Regional tilting

As already indicated, any tilts present are considered to be original and determined by the fabric of the country rock; there is no evidence of systematic post-emplacement tilting through most of the swarm. The Northern Granite, however, certainly caused local doming and this may have rotated directions of magnetization of earlier dykes in the area of NE Arran included in Tyrrell's bands 6 and 7. Dips and strikes marked on the geological map together with slopes estimated from Figure 21 of the Arran Memoir (Tyrrell 1928) indicate values of up to 30° towards ESE. If this movement has affected the dykes it would give normal magnetizations a slight westward bias in declination and reversed magnetizations a similar bias to the east. Fig. 8(a) shows that although the normal dykes of bands 6 and 7 do in fact have a pronounced westward bias the reversed dykes do not have a corresponding bias to the east, so 'correcting' both groups for this tilt would still leave a normal-reverse asymmetry.

Thus although there may be local effects in bands 6 and 7 which will be discussed again in Section 4.7, tilting is not a general problem.

4.2.2 Remagnetization

Remagnetization is also considered to be unlikely on any major scale. This can be adduced from various results: the internal consistency of directions of the four cores taken from each unit is very good and most secondary components of magnetization were removed by low fields (~ 10 mT) or low temperatures; some of these secondary components are normal, some are reversed; there are several instances of bodies of one polarity cutting others of the opposite polarity; there is no significant difference in direction between the dykes of differing petrography (see Section 4.7); and the results of af and thermal demagnetizations are in agreement. Further support comes from the thermomagnetic studies. Only five of the 128 Curie-point curves show evidence of two components; the majority of heating and cooling curves have similar simple geniculate form with a tendency for the final magnetization to be less than the initial magnetization. Although the Curie points range from 290 to 625°C, 70 per cent exceed 500°C with a mode between 500–600°C; the ranges and shapes of the distributions for the normally magnetized samples (73) and reversely magnetized samples (48) are similar.

As the dykes are now exposed at a level which was almost certainly deeply buried at the time of intrusion, the high Curie points probably result from a combination of deuteric and hydrothermal alteration which took place soon after intrusion as in the case of the Mull dykes (Ade-Hall, Palmer & Hubbard 1971).

The situation around the Northern Granite again needs closer consideration. The majority of dykes in bands 6 and 7, closest to the granite, are normally magnetized; so is the granite. However, first of all it must be pointed out that the direction deduced for the granite is quite different from the mean direction of bands 6 and 7, (Tables 2 and 3). Secondly the geological evidence (Richey 1961) suggests that the granite had little metamorphic effect on the country rocks more than a few hundred yards from the contact.

Evidence from individual dykes is inconclusive unless it is known whether or not they pre-date the granite and information on this point is not available. However, for the neighbouring dykes P538, P538A of band 7 the former has a single-component normal magnetization and a high (550°C) Curie point while the latter has a low (375°C) Curie point and is reversely magnetized with a large secondary component in the normal sense which was easily removed by heating to 100°C.

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Table 3. Mean directions of dykes for Tyrrell and Knaap bands.

		N	*R*	*D* (° ETN)	*I* (° down)	α (°)
			Tyrrell bands			
(1)	All	162	155.6	2.03	63.3	2.36
	N	27	26.2	357.8	61.9	5.0
	R	128	125.2	184.6	−65.3	1.9
(2)	All	107	102.5	356.2	64.1	2.9
	N	9	8.6	349.4	60.0	11.1
	R	88	85.9	180.1	−64.5	2.4
(3)	All	50	47.8	2.7	69.4	4.4
	N	–	–	–	–	–
	R	48	46.4	186.4	−68.5	3.9
(4)	All	52	50.0	354.5	67.9	4.0
	N	4	3.9	356.3	67.2	14.8
	R	46	44.7	175.4	−67.0	3.6
(5)	All	20	18.7	356.9	58.8	8.9
	N	–	–	–	–	–
	R	18	17.6	173.8	−62.2	5.1
(6)	All	25	24.5	347.3	61.5	4.4
	N	18	17.9	340.1	61.3	2.9
	R	7	6.7	186.1	−60.1	14.1
(7)	All	21	20.1	339.2	62.6	6.9
	N	15	14.9	330.8	62.9	3.1
	R	5	4.9	190.2	−68.3	14.4
			Knaap bands			
(a)	All	21	20.5	3.7	67.7	5.3
	N	2	2.0	346.2	71.9	9.9
	R	19	18.5	185.2	−67.1	5.8
(b)	All	33	32.2	2.1	64.7	4.0
	N	3	3.0	346.7	55.7	7.8
	R	29	28.5	183.1	−66.6	3.7
(c)	All	32	31.2	0.8	58.2	4.1
	N	8	7.8	2.3	51.8	8.5
	R	23	22.7	178.3	−61.6	3.7
(d)	All	32	29.3	359.0	64.3	7.8
	N	7	6.7	351.8	70.1	12.5
	R	22	21.5	187.1	−67.7	5.1
(e)	All	36	35.0	2.9	62.8	4.2
	N	7	6.9	3.6	64.5	8.9
	R	28	27.3	184.6	−63.1	4.6
(f)	All	49	46.9	1.4	66.5	4.3
	N	5	4.8	347.5	62.1	19.2
	R	39	38.1	184.8	−64.5	3.7
(g)	All	34	33.0	354.7	64.3	4.4
	N	3	2.9	349.1	53.1	24.4
	R	30	29.3	176.5	−66.5	4.0
(h)	All	18	17.0	347.2	62.3	8.8
	N	1	–	2.9	69.1	–
	R	13	12.8	177.5	−66.9	5.2
(i)	All	28	26.8	5.0	71.9	5.8
	N	–	–	–	–	–
	R	27	26.3	190.5	−71.4	4.6

Symbols as in Table 2.

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4.3 DISTRIBUTION OF DIRECTIONS

It can be seen (Fig. 2) that there is a considerable dispersion about the centroids of the normal and reversed distributions: even when 53 intermediate dykes (see Section 4.4) are excluded, the *mean* directions for 85 normal units has an angular standard-deviation (Wilson 1959) of 12.8° and that for 369 reversed units is 13.5° (Table 2). As has been discussed above there is little evidence for any of this dispersion arising from lack of precision in the individual mean directions or of it being the result of tectonic movements after magnetization or of secondary remagnetization and attention must be drawn to the similarity of these distributions to those found for the Mull dyke swarm (Ade-Hall *et al.* 1972). Post-magnetization conditions were unlikely to have been the same for the two swarms because of the different characteristics of the two centres and the difference in distance of the dykes from the respective centres. Therefore it is considered that, in both cases, these directions represent the variation of the geomagnetic field during extended periods of intrusion.

4.4 PROPORTIONS OF NORMAL, REVERSED AND INTERMEDIATE DIRECTION DYKES

Of the 435 dykes accepted for analysis 339 (78 per cent) are reversed, 73 (17 per cent) are normal and 23 (5 per cent) intermediate. Here an intermediate direction is defined as one with polar colatitude $\theta < 40°$ (Wilson *et al.* 1972b); the number would be reduced to 10 dykes if the criterion $\theta < 45°$ is used (Watkins 1973; McElhinny & Merrill 1975). The ratio R/(R + N) is 0.82. A similar preponderance of reversely magnetized units has been noted

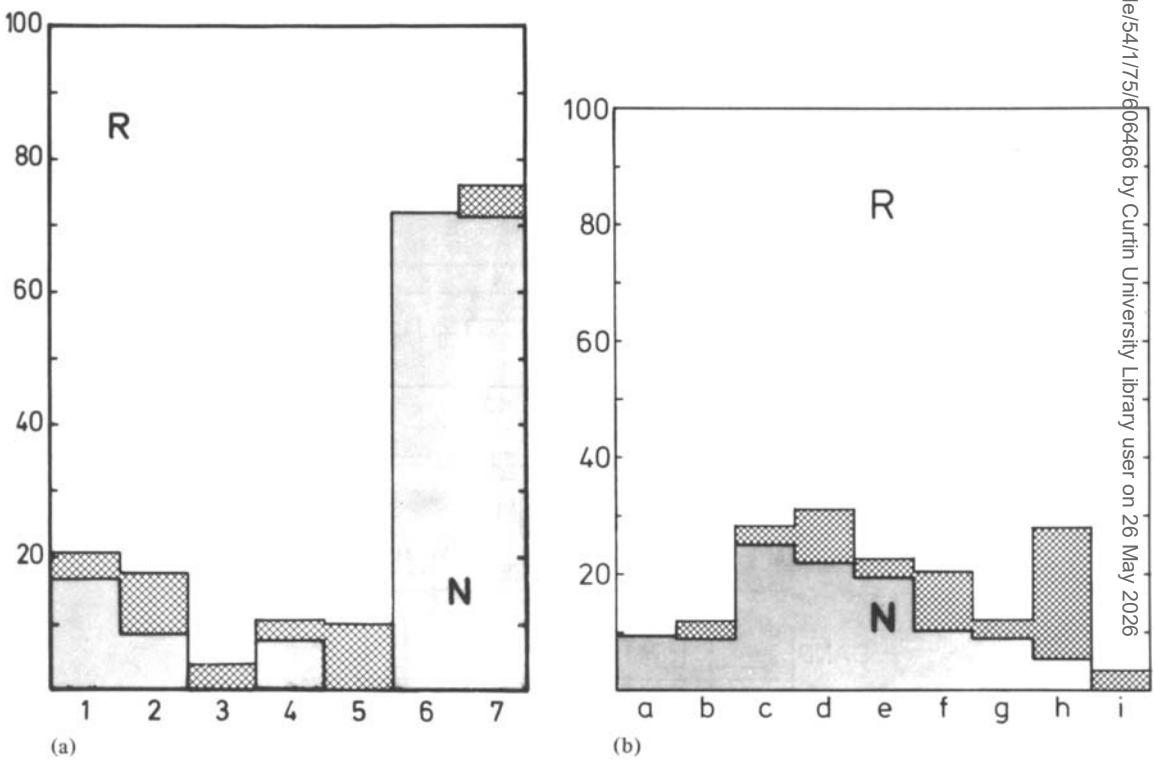


Figure 4. Percentage of normal, reversed and intermediate directions of magnetization for (a) Tyrrell bands, (b) Knaap bands.

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for other dyke swarms in the province: 0.92 in Lundy (Mussett *et al.* 1976), 0.73 in Mull (Ade-Hall *et al.* 1972) 0.81 in Skye (Wilson *et al.* 1974).

This mixture is not uniform along the Arran coast. The percentages of N, R and I directions of magnetizations for each of the Tyrrell bands given in Table 2 are plotted in Fig. 4(a). The most striking feature is the high percentage of normal dykes (~ 70 per cent) in bands 6 and 7 compared with the other bands (~ 10 per cent); there are no normal dykes in bands 3 and 5.

Knaap's bands, confined to the south coast, are narrower divisions and there is a high percentage of normal dykes in bands a, c, d and e with a general decrease from band d to i (Fig. 4(b)).

4.5 POLARITY AND TREND

Tyrrell (1928) noted that the modes for the various trends occurred in different bands. For instance bands 3, 4 and 5 contained no NE (337.5–056.25° E of N) trending dykes, whereas in bands 6 and 7 ~ 9 and 11.5 per cent of the dykes respectively had this trend. NNE trends also increase in bands 6 and 7 compared to 1–5; NNW dykes peak in band 2 and also increase in bands 6 and 7.

Knaap (1973) plotted 'rose-diagrams' to show the distribution of trends in each of his bands a–i which encompass the S Arran swarm, between Port-na-Feannaiche and Dippin Head. Over all these bands the predominant trend is 340–350°, i.e. north of NNW, whereas the SE Arran swarm between Dippin Head and Kingscross point has a predominant trend of

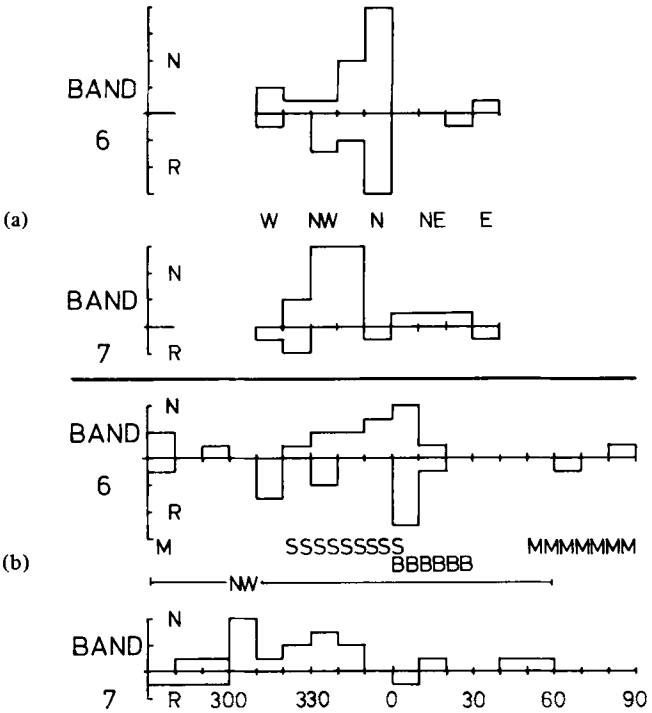


Figure 5. (a) Histograms showing number of dykes as a function of trend for each polarity in Tyrrell bands 6 and 7. (b) Same information subdivided to show the relationship to the trends of the separate swarms: S – S Arran swarm; B – Bute subswarm; M – Merkland subswarm; NW – NW Arran swarm.

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320–330° – i.e. more northwesterly. Within the S Arran swarm, Knaap bands c, d and e show a greater NW trend than the rest.

Superficially these distributions of trend together with the distribution of polarities noted above suggest a link between polarity and trend particularly in the northeast of the island. However, closer examination does not confirm this.

Although in the present study dyke trends were not measured in the field, it is possible to read some approximate values from the aerial photographs. The histograms of Fig. 5 display the results for bands 6 and 7 and show that there is no clear correlation of polarity with trend. It would appear that the correlation is with location rather than trend.

Knaap has suggested that some of the dykes in NE Arran may belong to the NW Arran dyke swarm and the Bute sub-swarm rather than the main S Arran swarm. The predominant trend of the NW Arran swarm is between 300–310° (although there is a wide spread from 270 to 060°) and that of the Bute sub-swarm 50–100°. The latter directions are present in band 6 but with no preferred polarity, while for band 7 the former trend is represented only by normal dykes. It could be that the high percentage of normal dykes in NE Arran is linked to the presence of the NW Arran swarm and further studies are planned to examine this possibility.

4.6 POLARITY AND THICKNESS

It was possible to record thicknesses for 343 dykes which ranged from 0.18 to 30.0 m. The average thickness for these is 2.25 ± 2.2 m (standard deviation) compared with Knaap's values of 2.2 m for the S Arran swarm, derived from the dependence of percentage dilation on the number of dykes per kilometre and 2.6 m for the whole of the Islay, Jura and Arran swarms. The mean thickness of 58 normal dykes is 1.8 ± 1.3 m (0.17 m standard error of the mean), of 204 reversed dykes it is 2.4 ± 2.2 m (0.10 m standard error) and of 23 intermediate dykes 1.22 ± 0.71 m (0.15 m standard error). The normalized histograms (Fig. 6) show that the difference of average thickness between the normal and reversed dykes could

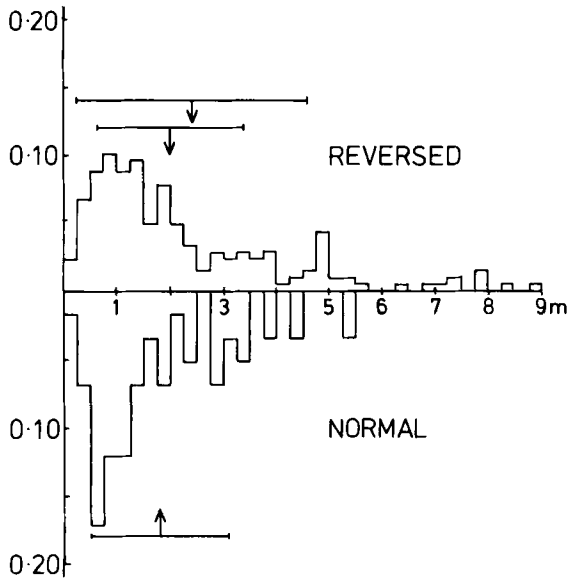


Figure 6. Normalized histograms of dyke thickness for normal and reversed polarity groups. The mean values and standard deviations discussed in the text are indicated.

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be largely due to the 14 reversed dykes greater than 6 m wide. Excluding these, the mean thickness for the remaining 190 reversed dykes is 2.0 ± 1.4 m (0.10 m standard error) which does not differ significantly from the normal mean. The 14 thick dykes do not appear to have any other characteristic in common except that several belong to multiple or composite dykes.

4.7 POLARITY AND PETROGRAPHY

A simple preliminary classification was made between those dykes definitely containing olivine which we attributed to the alkaline–olivine–dolerite group (Tyrrell 1928) and those with no olivine, attributed to the quartz–dolerite–tholeiite group. One hundred and forty-four of each group were examined. The dykes containing olivine were made up of 20 normal, 122 reversed and two intermediate compared with 21 N, 117 R and 6 I for those without olivine, so that there is no simple correlation of polarity with petrography.

4.8 GROUP MEAN DIRECTIONS

4.8.1 Dykes

The overall mean direction for the 435 accepted dykes is 357.6° E, +64.5° (down), α_{95} = 1.4° which is not consistent with the direction of a centred axial-dipole field (0° E, +71° down) at this latitude (55.5° N, longitude 354.8° E). Exclusion of the 23 intermediate directions does not change the mean significantly. The 73 normally-magnetized dykes have a mean direction 346.7° E, +62.4°, α_{95} = 2.2° and the 339 reversely-magnetized dykes have a mean direction 181.9° E, −65.6°, α_{95} = 1.3°. These directions are displayed in Fig. 7 and

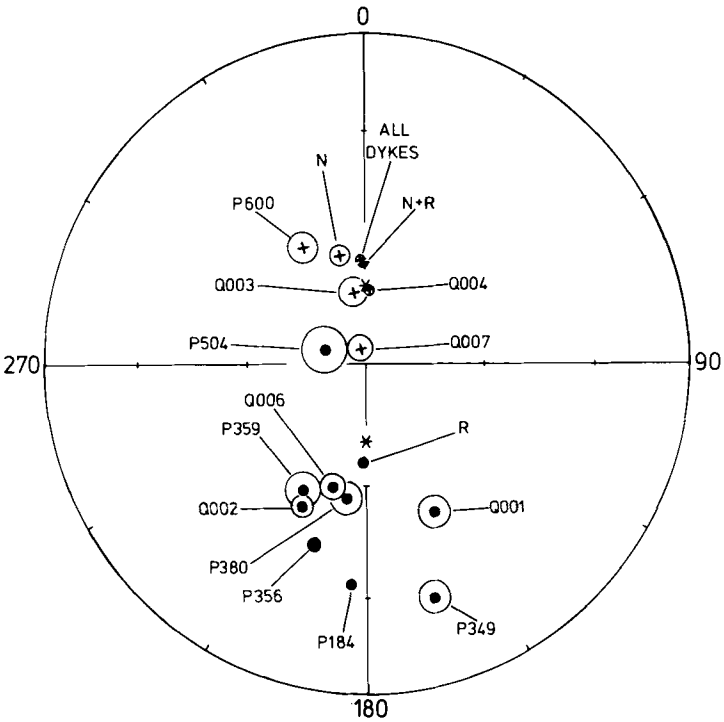


Figure 7. Mean directions of magnetization for dykes (all, N, R and N + R), sheets, sills and granite. Equal-area projection: + – lower hemisphere, inclinations positive; • – upper hemisphere, inclinations negative; * – centred axial-dipole field.

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Table 2. It is clear that the means of neither group match the axial-dipole field, though the reversed group is closer. Both groups in fact have a mean direction which is shallower than the expected dipole field. Similar departures from symmetry have been observed with other Tertiary data, consistent with a northward displaced axial dipole (Wilson 1971), or with a northward drift of Scotland at a rate of about 2 cm/yr during the last 60 Myr, or a combination of both.

As implied above, the normal and reversed means are not antiparallel; the reversed mean is almost due south but the normal mean lies to the west of north. A similar asymmetry between normal and reversed mean directions was found for the Lundy dykes (Mussett *et al.* 1976). This general pattern is not unlike that found for the Mull dyke swarm (Ade-Hall *et al.* 1972), although for Mull the mean normal direction was steeper than that of a centred axial dipole.

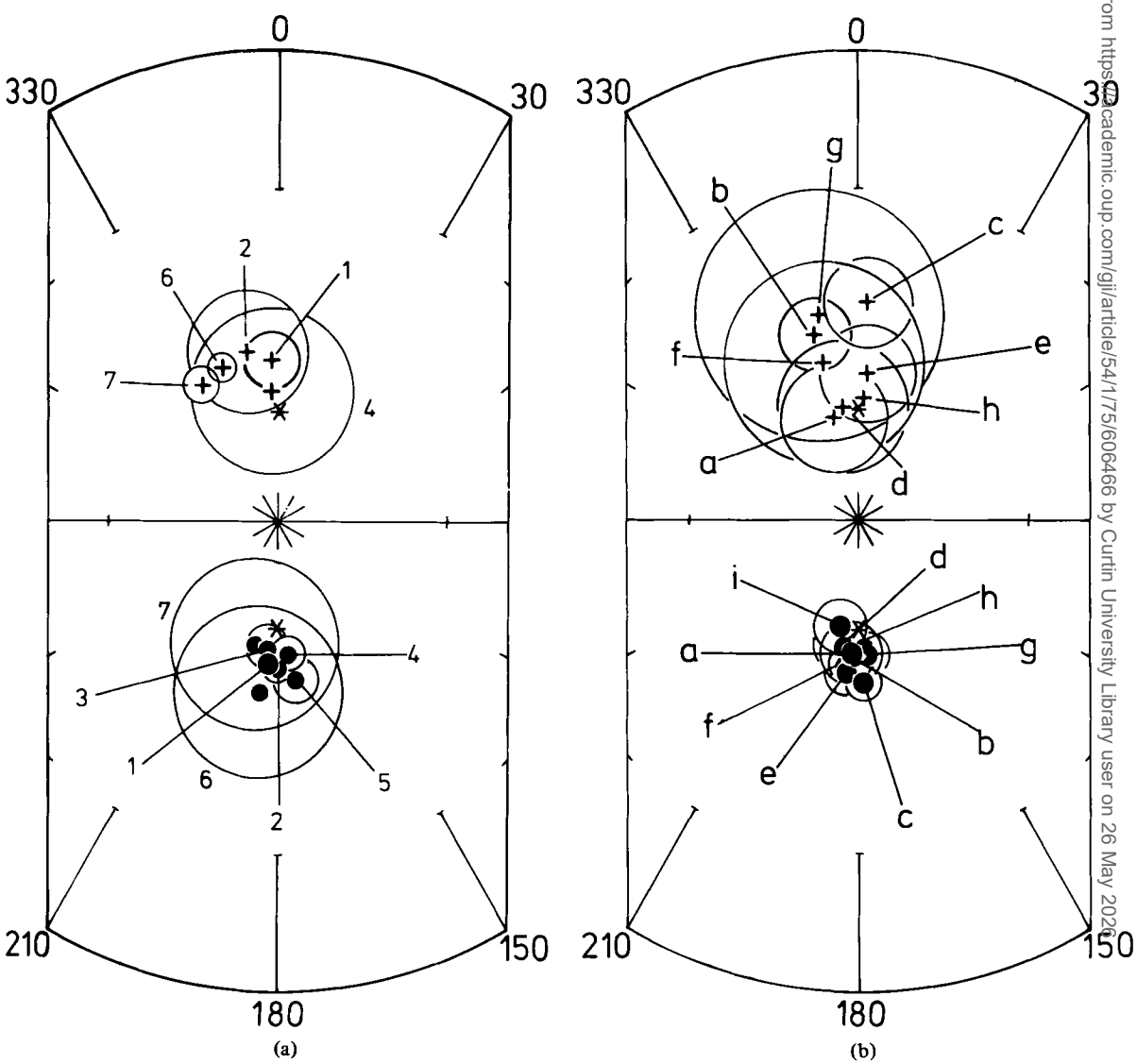


Figure 8. Mean directions of magnetization for normal and reversed groups in each division: (a) Tyrrell bands; (b) Knaap bands. Symbols as in Fig. 7.

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The mean overall, mean normal and mean reversed directions of magnetization for the dykes in each Tyrrell band (Table 3) are plotted in Fig. 8(a). The mean reversed directions show little change from band to band; on the other hand the mean normal direction for bands 6 and 7 do appear to be different from the normal means for the other bands. The asymmetry between N and R directions is different for the various bands which suggests that the groups may not all be members of the same population. If a tilt correction (4.2.1) was applied to the normal directions alone they would have more northerly direction and the asymmetry would be less. It could be that the normal dykes are older than the granite and have been tilted whereas the reversed dykes in these bands are later and unaffected.

The difference between the normal and reversed axes is also apparent within the smaller divisions of Knaap; the mean normal directions of all his bands (Table 3) except c, e and h are $\sim 10^\circ$ west of north whereas all the mean reversed directions cluster closely about the southerly direction (Fig. 8(b)).

However, there does not appear to be any statistical difference between the overall mean directions for the two petrographic groups (without olivine 357.3°, $+62.75$, $\alpha = 2.21$; with olivine 1.4°, $+64.4$, $\alpha = 1.94^\circ$).

4.8.2 Other intrusives

Those units which are identified as sills or granites have been considered separately; their unit mean directions are given in Fig. 7 and Table 2. Of the 13 units, four are normal, seven reversed and two intermediate (polar colatitude $< 40^\circ$, Wilson, Dagley & McCormack 1972b). The four normal units appear to define three different directions. The quartz–porphyry sills (Q003, Q004) have a direction consistent with a centred axial-dipole field; the granite Q007 (older, coarser type) has a considerably steeper direction while the Holy Island sill (P600) is shallower. The reversed units also show a variety of remanent directions, none of which like close to any of the axes defined by the normal units, but all of which are shallower than the mean direction of the reversely-magnetized dykes.

5 Age and magneto-stratigraphy

The evidence of intersecting intrusions can be used to establish a time sequence. Sixty-seven clear relationships were noted. Of these 55 are cases where the two intersecting bodies have the same polarity (1NN; 54RR) but there are five examples of younger normally-magnetized dykes cutting older reversed dykes and three cases where the reversed body is younger (in the remaining pairs at least one dyke failed to yield a clear polarity). The simplest polarity sequence consistent with these findings is either R → N → R or N → R → N.

It is also generally considered (Tyrrell 1928; Richey 1961) that the Brown Head and Bennan Head quartz–porphyry sills (Q003, Q004) are later than the crinanite sills (Q001, Q002, Q005, Q006). Since the former are normally magnetized and the latter reversely magnetized there must have been a R → N transition. The crinanite sills are believed to be contemporary with the inferred plateau-lava period which was early in the sequence and almost certainly accompanied by dyke intrusion so that the sequence R → N → R is preferred. The early dykes, lavas and crinanite sills would be associated with the first reversed interval, the quartz–prophyry sills and some dykes with the normal interval and later dykes with the second reversed interval.

This simple pattern might also be consistent with the succession suggested by Tyrrell (1928); he points out that the alkaline–olivine–dolerite dykes do not cut the Northern Granite or the Central Ring Complex whereas the quartz–dolerite–tholeiite dykes cut both

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these bodies. The Northern Granite (Q007, Q008) is normally magnetized and in spite of its different mean direction could belong to the same interval as the few normally-magnetized dykes of both petrographic types, while the majority of dykes of both types could fall in the earlier (alkaline–olivine–dolerite) and later (quartz–dolerite–tholeiite) reversed intervals.

Fourteen of the freshest dykes (26 or better on the classification of Mussett *et al.* 1973) gave a spread of ages from about 53 to 69 Ma. But as many lavas in the British Tertiary igneous province give ages which are not stratigraphically consistent (e.g. Mussett *et al.* 1973; Evans, Fitch & Miller 1973) and as one dyke from Mull gave a clearly excessive date (Mussett *et al.* 1973) little weight will be attached to these results in the absence of independent evidence for their validity.

The Holy Island sill (P600) is considered to be later than most of the dykes and as its polarity is normal this implies a late R → N sequence giving a polarity sequence for Arran of R → N → R → N. Its late stratigraphic position derives from Tyrrell's (1928) observation that though it cuts dykes, no dykes cut it. Knapp (1973), however, reported that some dykes do cut it so that it may not be as late as previously supposed and therefore it is not inconceivable that it belongs to the same polarity interval as the Northern Granite and/or the quartz–porphyry sills, so involving only the shorter sequence R → N → R. However, both the different paleomagnetic directions and the different radiometric dates (Northern Granite, 58.8 ± 0.6 Ma, Evans *et al.* (1973); Holy Island, two dates of 56 and 54.9, reported by Macintyre (1973) (one date reported by Macintyre used different decay constants; using the usual values reduces age to the 54.9 Ma quoted)) favour them belonging to different periods.

Although we cannot tell whether one or more normal intervals are represented, since the number of normally magnetized bodies is small the total time for which the geomagnetic field had normal polarity was probably quite short. This conjecture is supported by the magnetic polarity timescale of La Brecque, Kent & Cande (1977) which shows normal intervals $\leqslant \sim 0.5$ Ma long at this time. The Northern Granite could be associated with anomaly 25 (58.67–59.16 Ma) and Holy Island with the double anomaly 24 (55.58–55.81 and 56.11–56.60 Ma) with the whole igneous episode spanning ~ 3 Ma.

This period though still relatively short, is adequate for possible plate motions, changes in the non-dipole field and secular variation of the main geomagnetic field to give the large dispersion of directions of magnetization observed.

It should be noted that all the magnetic polarity timescales, including that of La Brecque *et al.* (1977), are still subject to possible errors of a few million years at this age, so that the absolute age of any particular anomaly is uncertain. However, this uncertainty does not affect the general conclusion that the whole episode is short.

The timescale proposed by Tarling & Mitchell (1976) with its widely separated and longer normal periods (55.02–55.84 and 59.20–60.05 Ma) appears to be less consistent with the data.

6 Conclusions

Of the 435 dykes analysed 78 per cent are reversely magnetized, 17 per cent are normally magnetized, while 5 per cent have intermediate directions. The directions appear to be original, with little remagnetization. The polarities do not correlate with thickness or petrography, but there is some correlation with location which may be because the axis of maximum density of intrusion shifted with time. A particular example of this is the predominance of normally-magnetized dykes along the northeast coast, which may belong to

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the NW Arran swarm. This concentration may be associated with changes in the regional stress field caused by the intrusion of the Northern Granite.

The Northern Granite and some of the sills are normally magnetized. Cutting relationships among the dykes establish that at least two reversals of the magnetic field occurred, and the simplest sequence of polarity compatible with the results is R → N → R, but if the Holy Island sill is truly late in the succession this needs to be extended to R → N → R → N. Since the Tertiary igneous rocks of Arran were formed *c.* 58 Ma ago the magnetic polarity timescale shows that if three polarity periods are involved the duration of activity lasted between 0.5 and 3.4 Ma; if four periods are involved the figures are 3.4 and 4.4 Ma.

The mean directions of magnetization of the reversed and normally-magnetized dykes are both shallower than the present-day centred axial-dipole field. This could reflect drift of Arran northwards since it formed, but since other parts of the British Tertiary igneous province also show departures but in the opposite sense (Mull normal dykes, Ade-Hall *et al.* 1972) changes in the geomagnetic field seem more likely.

Similarities with other parts of the province are reflected in the predominance of reversely-magnetized dykes, and in the presence of normally-magnetized centres (Skye: Brown & Mussett 1976; Mull: Bullerwell 1972, confirmed by unpublished palaeomagnetic measurements of this laboratory). This similarity can be only partly explained by contemporaneity because radiometric dates show that the Lundy dykes and Skye Red Hills are significantly younger than the Arran and Mull centres and other parts of the province.

Further work is in progress to refine the magneto-stratigraphy of Arran.

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